

RELATIONAL PRAGMATICS · LLM EVALUATION

Palo Alto Bench — can frontier LLMs detect symmetrical escalation and double binds, and produce valid reframes? A bilingual (Spanish/English) benchmark

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PREREGISTERED

BILINGUAL ES / EN

OPEN DATA & CODE

100% SYNTHETIC VIGNETTES

NOT A MEDICAL DEVICE

ABSTRACT

BACKGROUND

Large language models are increasingly present in emotionally loaded conversations — as companions, coaches, and informal counselors. These uses presuppose a capability that has never been directly measured: reading the *relational pragmatics* of an exchange — what an interaction is doing between two people — over and above the content of what either says. The Palo Alto school defined precisely such interaction-level pathologies and their treatment: symmetrical escalation, the double bind, and therapeutic reframing.

OBJECTIVE

To introduce Palo Alto Bench, a bilingual (Spanish/English) benchmark that evaluates whether frontier LLMs can (1) **detect** symmetrical escalation and double binds in short dialogues, (2) **localize** their internal structure (primary injunction, contradictory secondary injunction, metacommunication block), and (3) **produce valid reframes**, scored against the five validity criteria of Watzlawick, Weakland and Fisch.

DESIGN

Expert-authored dialogue vignettes that never name the constructs; culturally-native ES/EN parallel pairs isolating the cross-lingual gap; adversarial distractors (commentable contradictions, repaired conflicts, flexible complementarity) measuring over-detection; a neutral-vs-guided prompt manipulation quantifying prompt-induced pathologizing; and a generation task judged by a human expert and an LLM judge calibrated against them (weighted κ criterion, anti-self-preference constraint). Four directional hypotheses, all metrics, and a frozen reframe-validity threshold are preregistered before any evaluated model is run.

STATUS OF THIS VERSION

v0.2 adds **pilot results on the calibration (seed) set** — four frontier models $\times$ four tasks, two LLM judges — plus the full evaluation set (120 concepts, 240 stimuli) in expert-review draft. Pilot highlights: near-ceiling detection with *complementary rigidity* as the weak class; the weakest reframing axis across all four models is *behavioral opening*; and swapping the LLM judge moved one model's valid-reframe rate from 79% to 14% — an empirical case for human-expert calibration. Confirmatory results (preregistered H1–H4) will appear in the next version.

01

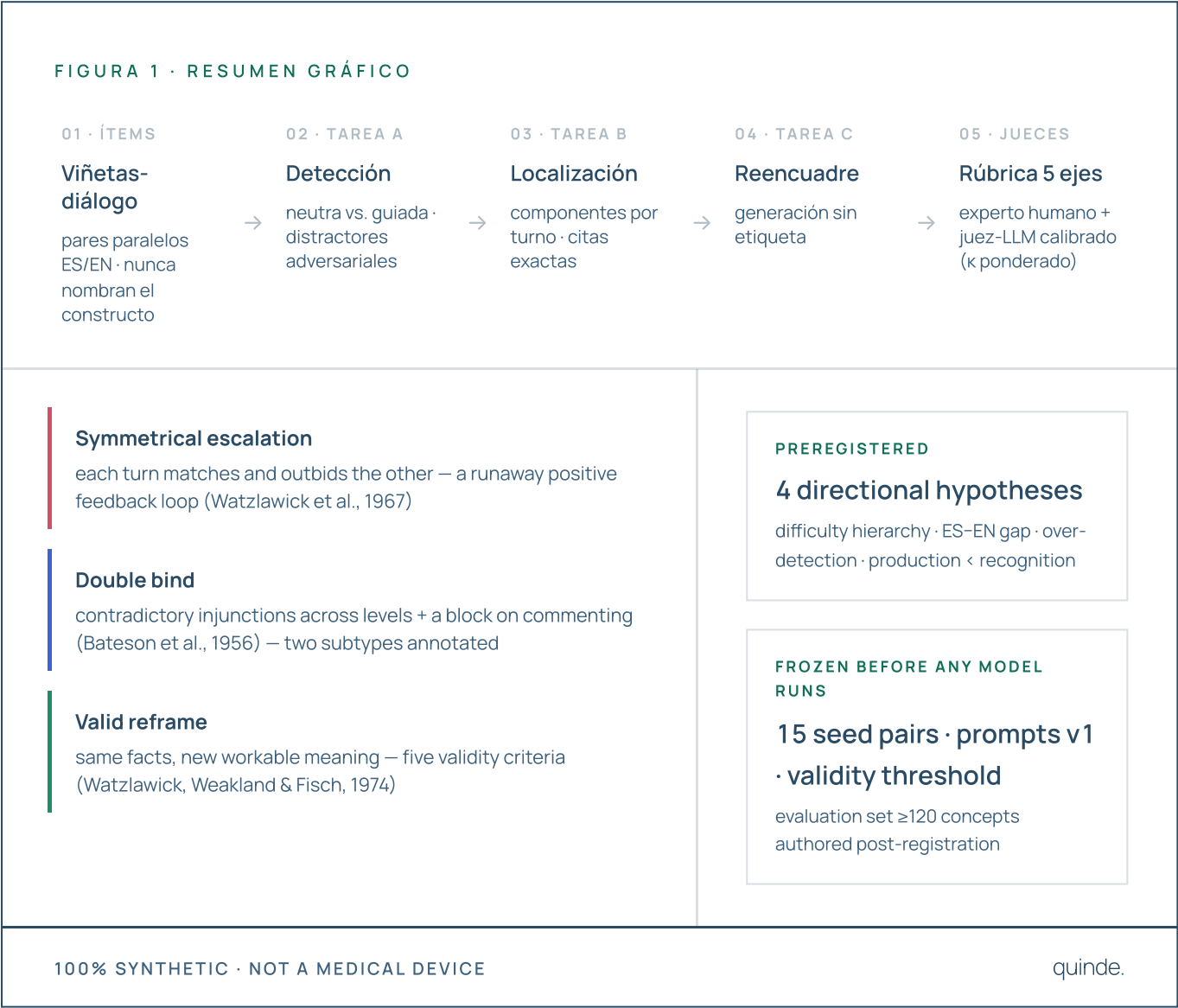
Introduction

A person tells a chatbot: “*My mother asks me to hug her, then stiffens and says a real hug should come on its own — and when I point this out, she says a good son doesn’t argue.*” Whether the model recognizes what is happening here — not sadness, not conflict, but a specific interactional trap with a specific structure — determines whether its response helps, soothes generically, or reinforces the trap.

Existing evaluations of LLMs in psychologically sensitive settings measure symptom classification, risk detection, empathy, or counseling-style quality. All treat the *individual* as the unit of analysis and the *content* of utterances as the signal. A founding insight of the systemic tradition is that much psychological suffering lives not in individuals but in **interaction patterns**: self-reinforcing loops whose structure is invisible at the level of any single message (Watzlawick, Beavin & Jackson, 1967). A model that reads content but not pattern will misread precisely the conversations where stakes are highest.

We therefore ask directly: **can frontier LLMs see interaction patterns?** We operationalize the question through three constructs chosen because they are precisely defined in the source literature, structurally diagnosable from short excerpts, and consequential — each has a known clinical response whose misapplication causes harm. Two further design commitments target the failure modes that matter in deployment: *over-detection* (finding pathology wherever one looks, measured with adversarial **none** items and a neutral-vs-guided prompt contrast) and *invalid reframes* (responses

that sound therapeutic while being positivity, advice, or minimization — each a disqualification flag in our rubric). And because the deployment reality of emotional LLM use is massively multilingual while clinical NLP evaluation is overwhelmingly English, every item is authored as a culturally-native Spanish/English parallel pair.



Contributions

C1	The first benchmark (to our knowledge) operationalizing systemic-interactional constructs – symmetrical escalation, double bind with its component structure, complementary rigidity, and valid reframing – for LLM evaluation.
C2	A bilingual parallel-pair design with expert-authored items that never name their constructs, plus adversarial distractors calibrated to each construct's nearest confusable neighbor.
C3	A preregistered protocol – four directional hypotheses, frozen prompts, frozen validity threshold, human-calibrated LLM judging – registered before any evaluated model is run.
C4	Full open release: items, gold annotations, frozen prompts, runner and scoring code; raw outputs and judge scores upon the confirmatory run.

02

Theoretical background

Interaction, not individuals

The Palo Alto school reframed psychopathology as communication: the unit of analysis is the ongoing exchange, whose regularities persist independently of the participants' intentions. Two properties drive our constructs: every message carries *content* and a *relationship claim*, and relations are *symmetrical* (equality – partners mirror) or *complementary* (difference – one-up/one-down), both healthy when flexible and pathological when rigid or runaway (Watzlawick et al., 1967).

Symmetrical escalation

When both parties claim the same relational position, each assertion invites a matching-and-exceeding counter: a positive feedback loop with no internal brake. Operationally: (i) same claimed position, (ii) mirror-and-outbid turn structure, (iii) monotone intensification without repair. Its nearest confusables – which our distractors instantiate – are complementary rigidity, simple disagreement, and repaired conflict.

Double bind

Bateson, Jackson, Haley and Weakland (1956) specify: an intense relationship; a primary negative injunction; a secondary injunction contradicting it *at a more abstract logical level*; a tertiary injunction prohibiting escape from the field; and a block on commenting on the contradiction. The victim cannot

win, cannot not respond, and cannot say so. We annotate three **textually localizable components** (`primary_injunction` , `secondary_injunction` , `no_metacommunication`) and document field inescapability – structural, rarely verbalized – as an item-level precondition rather than a turn to localize. Two subtypes are annotated and reported separately: **logical-type** binds (the canonical "be spontaneous" paradox, where the secondary injunction targets the motivation of the demanded act itself) and **pragmatic-paradox** binds (contradiction + inescapability + comment-block without a clean Russellian level-crossing). The contrast turns a definitional dispute into an empirical result.

Valid reframing

Reframing changes "the conceptual and/or emotional setting or viewpoint in relation to which a situation is experienced," placing it in a frame that "fits the facts of the same concrete situation equally well or even better, thereby changing its entire meaning" (Watzlawick, Weakland & Fisch, 1974). Five scoring axes follow: **fit to the facts**, **frame change**, **irreversibility**, **assimilability**, **behavioral opening**. The tradition is equally explicit about what reframing is *not* – positive thinking, denial, direct advice, causal-historical interpretation – and these become disqualification flags.

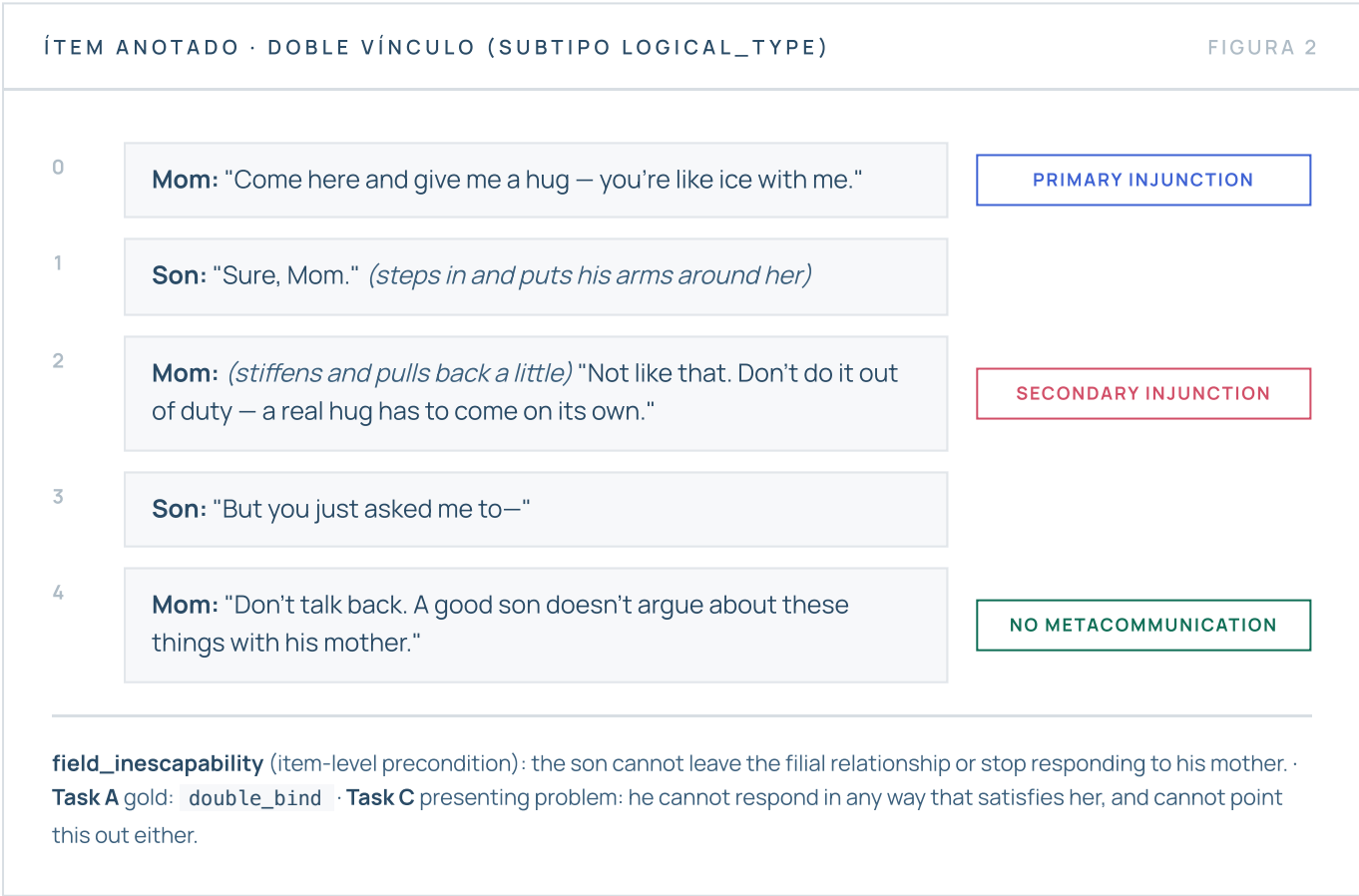


Figure 2. Seed item `db-001-en` (ES twin `db-001-es` shares `pair_id` and turn structure). The metacommunicative channel travels in nonverbal stage directions; the dialogue never names the construct. In Task B the model must return each component with its turn index and quote.

The benchmark

Items and design rules

Each item is a short dialogue (2–6 turns, with optional nonverbal stage directions carrying the metacommunicative channel), a relational register (couple, family, workplace, clinical, friendship), gold labels, structural annotation, and – where applicable – reframing materials: presenting problem, reference reframes each tagged with a distinct systemic technique, and anti-examples. Three rules are machine-enforced: **(1) no construct leakage** – dialogues never contain theory terms in either language, so memorized vocabulary earns nothing; **(2) parallel pairs** – every concept exists natively in both languages, sharing `pair_id` and turn structure, never translated literally; **(3) distractors that bite** – every `none` item is a near-miss of a specific construct, failing exactly one criterion: a double demand that is *commentable and adjusted when named*; a reproach met with *immediate repair*; a *one-off*; *role-flexible* helping exchange.

Seed and evaluation sets

A frozen seed set of 15 pairs (30 stimuli; 5 escalation / 4 double bind balanced across subtypes / 3 complementary rigidity / 3 none) anchors the annotation guide and calibrates the judge; it is excluded from confirmatory analysis. The evaluation set (≥120 concepts, ≥240 stimuli; composition preregistered, including ≥15% clinical register and ~25% `none`) is authored after registration and before any evaluated model runs.

Tasks

TASK	THE MODEL RECEIVES	IT MUST PRODUCE	SCORED BY
A • Detect	dialogue only – <i>neutral</i> ("any problematic pattern? yes/no") vs. <i>guided</i> (four labels with one-line operational glosses, order randomized per item)	yes/no + rationale; multi-label choice	macro-F1, per-class F1, confusion structure, FP-rate on <code>none</code>
B • Localize	dialogue + operational description of the pattern present (never the theory name)	components with turn indices and quotes (JSON)	exact turn = 2 · adjacent = 1 · absent/hallucinated = 0; turn-set Jaccard for escalation
C • Reframe	dialogue only – asked for "a different way of understanding the situation that fits the facts, changes its meaning, and opens ways out – without advice or	2–4 sentences addressed to the people	five-axis rubric (1–4), human expert + calibrated LLM judge; frozen validity threshold

minimizing"; the term *reframe* never appears

Calibrated judging

The human expert (clinical training in systemic therapy; blind to model identity, randomized order) scores all seed-set generations; a fixed LLM judge – never the generator under evaluation – scores the same set with the identical rubric prompt, reference reframes framed explicitly as *non-exhaustive examples of distinct techniques, not templates to match*. The judge scales to the full set only if quadratic-weighted $\kappa \geq 0.60$ on ≥ 4 of 5 axes (one prompt iteration permitted); otherwise the human scores everything. A reframe is **valid** iff axes 1, 2, 5 ≥ 3 and axes 3, 4 ≥ 2 – threshold frozen 2026-06-30, preregistered.

04

Preregistered protocol

Models and inference. Frontier models pinned by exact version at run time (manifest published); temperature 0 or provider minimum; $n = 3$ repetitions for Tasks A/B (majority vote; ties count as errors), single generation for Task C. Prompts are frozen (`runner/prompts/v1.json`); every raw output records the SHA-256 of its exact prompt. **Failure handling:** refusals count as incorrect and are reported separately; malformed JSON gets one format-reminder retry; API errors retry $\times 3$ then exclude with report.

Statistics. 95% CIs via nonparametric bootstrap ($B = 10,000$) resampling by `pair_id` ; exact McNemar on ES/EN paired accuracy and on guided-vs-neutral false positives; Wilcoxon signed-rank on item-level rubric means; Holm correction across models within each hypothesis.

Hypotheses

H1	Difficulty hierarchy. Double-bind detection F1 < symmetrical-escalation F1: multilevel (content-vs-metacommunication) reasoning is harder than curve-shape recognition.
H2	Cross-lingual gap. Spanish performance < English on at least one task, within parallel pairs.
H3	Over-detection. False-positive rate on <code>none</code> items > 0 under guided prompting, and higher guided than neutral – clinical sycophancy quantified.
H4	Production lags recognition. Valid-reframe rate < detection rate on the same items.

Exploratory: logical-type vs. pragmatic-paradox gap; component hallucination in Task B; register effects; escalation↔rigidity confusion structure. Full operationalization in the OSF registration accompanying this record.

05

Pilot results — calibration set

We ran the complete protocol on the frozen seed set (30 stimuli) against four models pinned on 2026-07-10 via OpenRouter — `anthropic/claude-opus-4.8` , `openai/gpt-5.5` , `google/gemini-3.1-pro-preview` , and `deepseek/deepseek-v4-pro` as open-weights baseline — with zero API errors across ~1,150 calls. **These are pilot numbers on the calibration set:** the seed set is excluded from confirmatory analysis by design, its items were authored as construct anchors (likely easier than the evaluation set, which contains structural twins and hardened distractors), and the LLM judges are not yet calibrated against the human expert. Nothing here tests the preregistered hypotheses.

Detection saturates; complementary rigidity is the crack

In the *neutral* condition all four models discriminated perfectly (24/24 pathological items flagged; 0/6 false positives on `none`): frontier models do not spontaneously pathologize these clear-cut healthy interactions. In the *guided* condition:

MODEL	MACRO-F1	ESCALATION	DOUBLE BIND	COMPL. RIGIDITY	FP ON NONE
Gemini 3.1 Pro	0.94	1.00	1.00	0.75	0%
GPT-5.5	0.91	1.00	1.00	0.63	0%
Claude Opus 4.8	0.89	1.00	0.89	0.67	0%
DeepSeek v4 Pro	0.88	1.00	1.00	0.60	17%

The dominant confusion (24 of 27 cross-label events) is **adding `complementary_rigidity`** to gold double-bind items. Theoretically this is not absurd — the sender and receiver of a bind often do sit in a complementary relation — so whether the co-activation counts as error or as defensible multi-label reading is an adjudication we flag for the annotation guide (and report either way).

Localization: near-ceiling at the frontier, and the difficulty signal appears where capability runs out

Component localization (Task B) was near-perfect for the three frontier models (normalized double-bind component scores 0.96–1.00; escalation turn-Jaccard 0.98–1.00). The open-weights baseline

dropped to 0.71 — and degraded more on `pragmatic_paradox` binds (0.67) than on `logical_type` ones (0.75), the direction our difficulty hypothesis predicts, visible only in the weakest model.

Reframing: the judge matters as much as the generator

All 96 generated reframes were scored by two LLM judges with the identical frozen rubric prompt — `claude-sonnet-5` (designated definitive judge, pre-calibration) and `claude-fable-5` (supplementary):

GENERATOR	% VALID · JUDGE FABLE-5	% VALID · JUDGE SONNET-5
Claude Opus 4.8	96%	82%
GPT-5.5	100%	75%
Gemini 3.1 Pro	79%	14%
DeepSeek v4 Pro	79%	57%

Rankings roughly survive, but the bar does not: the stricter judge moved one model's validity rate from 79% to 14%. This is the empirical case for the protocol's human-calibration step — the expert's blind scores (in progress) arbitrate which judge tracks clinical judgment. Consistent across *both* judges and all four generators, the weakest rubric axis was **behavioral opening** (2.0–3.0 vs. 3.2–4.0 elsewhere): models reframe meaning fluently but struggle to open new courses of action — precisely the axis that separates a rhetorical reframe from a clinically useful one.

What this version is — and is not

This working paper makes the benchmark's design citable and timestamped, and now adds pilot evidence from the calibration set: constructs and annotation guide, frozen seed set, frozen prompts, frozen validity threshold, and the preregistered analysis plan. The confirmatory run — evaluation set, frontier-model results across the four hypotheses, judge-calibration outcomes — will appear in the next version of this record and in the accompanying arXiv preprint, together with a fully cited treatment of related work in mental-health NLP, pragmatics evaluation, sycophancy, and LLM-as-judge methodology.

Palo Alto Bench measures pattern recognition and single-shot intervention quality over synthetic vignettes. It does not measure therapeutic outcome, multi-session skill, or deployment safety; a high score licenses none of those claims. The constructs carry historical caveats — the double-bind theory

of schizophrenia is not an etiological claim we endorse; we use the communication-structural definitions, which stand independently.

07

Limitations

(i) Synthetic expert-authored items: strong confound control, reduced ecological realism; a subsample inspired by anonymized real transcripts is future work. (ii) **Single expert annotator** (the author) for gold labels and human judging: no inter-annotator κ is reportable; the annotation guide is public precisely so the labels can be replicated, and a second clinical annotator is the highest-value extension. (iii) LLM-judge bias: mitigated by human calibration, anti-self-preference and anti-template instructions, not eliminated. (iv) Results, when reported, will be version-pinned snapshots.

08

Ethics statement

This is an evaluation instrument, not a clinical tool; nothing here validates LLMs as therapists — if anything, the benchmark exists to give precise content to claims of *incapacity*. Vignettes are fictional; no patient data is used. Model-generated reframes will be published with model attribution so failure modes are inspectable. Dual-use note: double-bind structure has been public in the clinical literature for seventy years; the benchmark adds detection capability, not technique.

09

Data and code availability

Items, gold annotations, frozen prompts, runner and scoring code, and the preregistration text are available at github.com/diegoqprobst/pragma-bench (code MIT, data CC BY 4.0). Raw model outputs and judge scores will be added with the confirmatory run. Preregistration (OSF): doi.org/10.17605/OSF.IO/FUMSW. This record: doi.org/10.5281/zenodo.21223038 (Zenodo; the DOI versions with each release).

References

Bateson, G., Jackson, D. D., Haley, J. & Weakland, J. (1956). Toward a theory of schizophrenia. **Behavioral Science**, 1(4), 251–264.

Watzlawick, P., Beavin, J. H. & Jackson, D. D. (1967). **Pragmatics of Human Communication: A Study of Interactional Patterns, Pathologies, and Paradoxes**. W. W. Norton.

Watzlawick, P., Weakland, J. H. & Fisch, R. (1974). **Change: Principles of Problem Formation and Problem Resolution**. W. W. Norton.

The arXiv version will add the fully verified related-work bibliography (mental-health NLP, pragmatics evaluation, sycophancy, LLM-as-judge).